THE UNIVERSITY OF WESTERN AUSTRALIA



CITS5206 Information Technology Capstone Project

Project Spec and Plans

Autonomous Mining Equipment Market Map

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Representative users

RCT Global · Epiroc

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Project and Resources

Client	Dr Adrian Boeing, UWA
Facilitator	Fudong Qin
Representative users	RCT Global (Yash Singh) · Epiroc (Milan Ivovic)

The client and the facilitator have access to every resource listed below.

Resource	Link
GitHub repository	https://github.com/zinken7/capstone-group-5
Project board	https://github.com/zinken7/capstone-group-5/projects
Sprint milestones	https://github.com/zinken7/capstone-group-5/milestones
Meeting minutes	https://github.com/zinken7/capstone-group-5/tree/main/docs/meeting_minutes
Project plan	https://github.com/zinken7/capstone-group-5/blob/main/docs/project_plan.md
Software Requirements Specification	https://github.com/zinken7/capstone-group-5/blob/main/docs/SRS/SRS.md
Database design (ERD)	https://github.com/zinken7/capstone-group-5/blob/main/docs/database/ERD.md
Correspondence log	https://github.com/zinken7/capstone-group-5/blob/main/docs/correspondence-log.md
MS Teams, Group 5 channel	Group 5

1 Problem Statement and Key Deliverables

1.1 Client and Context

The client is Dr Adrian Boeing of UWA. Two industry representative users have also contributed to the requirements: Yash Singh of RCT Global, working in market data analytics, and Milan Ivovic, Regional Business Manager for Asia Pacific at Epiroc. Both participated in the client and representative-user meeting on 11 August 2026 [1].

Their involvement helps ensure that the project addresses both the client's goal of creating an open dataset and the practical needs of industry users who use market information to understand technology adoption, competitor activity and potential opportunities.

1.2 The Problem

Information about autonomous and teleremote mining equipment is currently fragmented across company announcements, mine-site reports, industry publications and commercial market reports. Much of the more complete market data is commercially held. The client reported that relevant market reports can cost in the order of $50,000 to $100,000, placing them beyond many startups, smaller suppliers and academic researchers [2]. No comparable open dataset was found to exist [2].

This creates a practical problem for organisations trying to understand the market. Information can be difficult to obtain, slow to update and hard to independently verify. The representative users described cases where a supplier may only become aware of a competitor's presence after a large number of machines have already been deployed, or where changes in the automation technology used behind a machine brand may go unnoticed for months [1]. The usefulness of the data therefore depends not only on accuracy, but also on coverage and how quickly new information becomes visible.

Existing commercial dashboards do not fully solve this problem because their underlying data is not openly available for independent checking [1]. The project therefore needs to provide both a useful interface and an underlying dataset that users can inspect, contribute to and verify.

1.3 Why the Client Wants This Software

The client's aim is to make this type of market information more openly accessible. The requested system is an open, crowdsourced alternative to commercial market databases, with the structured dataset intended for publication under a Creative Commons licence [3].

For industry users, the value is practical. The system should make it easier to answer questions such as:

- where autonomous and teleremote equipment is being deployed;
- which manufacturers and technology suppliers are involved;
- how many machines are recorded at particular sites or in particular markets;
- how adoption differs by country, equipment type or automation category; and
- where competitor activity or potential opportunities may be emerging.

For smaller organisations and researchers, the open dataset removes the need to rely entirely on expensive commercial reports. It also makes individual records more transparent. Evidence can be attached to contributed information and each record carries a visible verification state, allowing users to distinguish between submitted and reviewed data.

Openness is therefore part of the solution rather than an additional feature. Data anyone can obtain gets reused in ways its publisher did not plan for, and that is where its value comes from; McKinsey put the annual value of open data across seven sectors, one of them oil and gas, at more than $3 trillion [4]. The extractive industries already publish on this basis, so that reported figures can be checked rather than taken on trust [5]. Open-source software follows the same pattern, most of its value going to those who reuse the code rather than to those who

wrote it [6]. A closed version of this system would leave the client's main concern untouched: that useful market information is hard to obtain and hard to verify.

1.4 Proposed Solution

The project will build a web-based, crowdsourced and queryable geospatial database of autonomous and teleremote mining equipment.

The system will combine four main user-facing functions:

1. an interactive GIS map of mine sites and recorded deployments;
2. an analytics dashboard for market-level summaries and trends;
3. a searchable and exportable equipment register; and
4. a contribution and verification workflow for adding and correcting information.

The underlying data model will connect mine sites, companies, machine models, manufacturers, technology products, technology suppliers and equipment deployments. Records can also include evidence and a verification status.

The system is intended as a market reference dataset and presentation layer. It does not perform mine planning, fleet dispatch or operational control.

1.5 Minimum Viable Product

The MVP is defined from the requirements currently prioritised **Must** in the Software Requirements Specification [7]. The status of client agreement to this scope is set out in Section 2.

#	Deliverable	MVP scope	Acceptance criteria
1	**Core data model**	Store companies, mine sites, machine models, technology products and equipment deployments. Manufacturers and technology suppliers are represented separately, with deployment and automation information stored in structured form.	FR-01 to FR-11
2	**Interactive GIS map**	Display one marker per mine site, distinguish deployment status, show deployment details and allow filtering by technology supplier.	FR-15 to FR-19
3	**Analytics dashboard**	Provide headline totals, cumulative machines over time, counts by country and machine type, supplier/manufacturer share and filters that apply to the displayed measures.	FR-21 to FR-28
4	**Equipment register**	Provide a searchable, filterable and sortable view of deployment records with tabular export.	FR-36, FR-37
5	**Contribution and verification**	Allow users to submit new information and supporting evidence. New records carry a verification status, while authorised administrators can review, correct, approve or reject submissions.	FR-40 to FR-46

Deliverables 2 to 5 correspond to the main user-facing functions requested by the client. Deliverable 1 provides the structured data foundation required by all four.

Because the dataset will initially cover only recorded deployments, analytics will describe the contents of the available dataset rather than claim to represent the complete global market. Coverage limitations will be made clear where appropriate.

1.6 MVP Boundary and Out of Scope

The MVP focuses on establishing a complete working flow:

collect data → review data → store structured records → explore the data through the map, register and analytics dashboard.

AI-assisted evidence checking and Retrieval-Augmented Generation (RAG) are possible extensions but are not dependencies of the MVP. The core system must remain usable without an LLM, automated web search or AI verification service.

Other features outside the MVP include self-service bulk upload, institutional AAF authentication, automatic machine counting from photographs, fleet-management or dispatch functionality, market forecasting and a dedicated mobile application [7]; Section 2.3 records the reason for each. These exclusions keep the project focused on collecting, maintaining and presenting structured market data within the available project timeframe.

2 Client Communication and MVP Agreement

The team treats client contact as the mechanism by which requirements are corrected, not as reporting. A written agenda or question set is prepared before each contact, and every meeting is recorded, minuted and committed to the repository, so that the client's own words remain the source for every requirement traced to them. The forward communication protocol, including meeting cadence and escalation, is set out in Section 3; this section records what has been communicated, what the team is proposing, and what remains unconfirmed.

2.1 Stakeholders

Commissioning the system and using it are different roles [2], and the requirements come from both.

Stakeholder	Interest in the project	How engaged
Dr Adrian Boeing (UWA), Client	Set the brief, commissioned the work and defines what success means; will use the dataset himself, as a researcher; sets the constraints on technology and licensing, and holds the system after handover	Meetings, email and MS Teams. At his request every group member is copied on correspondence rather than contact passing through one person [2]
Yash Singh (RCT Global), Market Data Analytics	Representative user. Works with the market data this system is meant to produce, and a source of both terminology and seed data	Requirements elicitation meeting, 11 August [1]
Milan Ivovic (Epiroc), Regional Business Manager, Asia Pacific	Representative user. Depends on market intelligence for commercial decisions in a region the dataset must cover	Requirements elicitation meeting, 11 August [1]
Fudong Qin (UWA), Unit Facilitator	Monitors progress on behalf of the unit, and advises on process and engineering practice	Fortnightly facilitator meetings; copied on client correspondence as evidence [8]
Group 15	Delivers the same brief in parallel, and shares both the client's time and the same elicitation meetings	Shared client meetings; the group is identified explicitly in all correspondence to avoid confusion between the two [2]
Teaching team	Unit administration and assessment	Dedicated unit and group channels in MS Teams; questions raised in person with the unit coordinator during class
Public / downstream users	The dataset is intended to be openly licensed, with CC BY 4.0 proposed for the data and MIT for the source code [7], so that others may reuse both. Startups, smaller suppliers and researchers priced out of commercial market reports are the intended beneficiaries [2]	Not engaged directly this semester. Their interests are carried by the client, who commissioned the system for that purpose, and tested against the representative users

Dr Boeing is both the commissioning party and a user of the dataset: as a researcher in autonomous mining equipment he has an interest in the technical approach as well as in the resulting data, and he nominated three existing systems as reference models for the map and the crowdsourced submission mechanism [2]. He also determines the constraints within which the system is built.

The two representative users were engaged because their roles impose different requirements. Yash Singh works in market data analytics, tracking market share monthly and product roadmaps quarterly; his requirement is coverage and recency, since he needs to see a competitor gaining ground early rather than after the fact [1]. Milan Ivovic holds commercial responsibility for a region; his requirement is depth, since he advises a mine operator by comparing their site with another where the same technology already works, and a record only serves that if it holds what makes two sites comparable: the commodity, whether the site is surface or underground, and the machines and technology installed [1]. Meeting them alongside the client in a single session shortened the process, since the client held a settled view of what he wanted while the representative users supplied the industry perspective against which it could be tested.

2.2 Communication Record

Date	Contact	Participants	Purpose and outcome	Evidence
23 Jul	First contact	Group 5 → client, all group members copied	The group requested a first meeting by email, with a link to its availability. A follow-up was sent on 28 July. The client replied on 31 July with a calendar invitation for an in-person meeting on 6 August, and undertook to arrange a separate meeting with an industry partner	Correspondence log
5 Aug	Question bank issued	Group 5 → client	Circulated in advance, structured around the four areas the project had to settle: the problem, the MVP boundary, ways of working, and technical constraints	`meeting_questions.md`
6 Aug	Client Meeting 1, in person	Dr Boeing, Group 5	Problem, users, data model and provisional MVP boundary; the team demonstrated a single-page prototype dashboard	CM1 minutes [2]
7 Aug	Facilitator Meeting 1, online	Fudong Qin, Group 5	Process expectations: define the MVP early and agree it with the client; minute every meeting and rotate the role; copy the facilitator on client correspondence as evidence	FM1 minutes [8]
11 Aug	Requirements elicitation, online	Dr Boeing, Yash Singh (RCT Global), Milan Ivovic (Epiroc), Group 5, Group 15	The team demonstrated the working prototype and the screen mockups, and took field-level and workflow requirements from the representative users against them. Scope decisions recorded in Section 2.3	CM2 minutes [1]
12 Aug	Specification, scope checklist and prototype issued	Group 5 → client, facilitator and all group members copied	Requirements specification, MVP scope checklist and the interactive prototype rebuilt from the meeting mockups (PR #4), sent by email and flagged on MS Teams, identifying the two answers most needed if time was short, the scope checklist and the UWA environment question, and offering to walk the client through either document. No response received	Appendix B of [7]; prototype; Correspondence log
13 Aug	Advice sought	Group 5 → unit coordinator and facilitator	Position set out and advice requested on whether the report could proceed on documented assumptions or whether a signed agreement on the MVP scope was required. Directed to continue following up with the client, and to prepare the report without waiting and revise it once confirmation arrives. A one-week extension was granted	Correspondence log

Date	Contact	Participants	Purpose and outcome	Evidence
14 Aug	Follow-up and progress update	Group 5 → client; Group 5 → unit channel	Confirmation of the scope checklist requested again by MS Teams, and a progress update posted to the unit channel	Correspondence log
18 Aug	Facilitator Meeting 2, online	Fudong Qin, Group 5	Progress reported, including the outstanding client confirmation. Facilitator directed the team to keep pursuing the client and settle the MVP early. Confirmed that the technology stack may be revised later if the reason and trade-off are documented	FM2 minutes [34]
19 Aug	Follow-up and offer of a call	Group 5 → client, facilitator and all group members copied	Reply sent on the email thread and by MS Teams, restating the request after the deadline extension and offering a call, with an availability link, so that the checklist could be worked through together	Correspondence log
25 Aug	Report issued for approval	Group 5 → client, facilitator and all group members copied	The submitted report sent into the same thread, with the MVP scope in Section 2.3 put forward for approval and a request that any item requiring change be identified	Correspondence log

Every record above is held in the project repository at https://github.com/zinken7/capstone-group-5. Each contact is recorded in the correspondence log at `docs/correspondence-log.md`, with its date, channel, recipients and outcome.

Every meeting is minuted from the recording, with actions, owners and dates, and the minute-taking role rotates as the facilitator directed [8]. Every requirement in the specification cites the contact it came from, so a claim about what the client wants can be checked against the source rather than taken on trust.

During a meeting the team listens first. A question bank is prepared beforehand, but the questions are held in reserve, since a stakeholder's own account answers many of them. The bank then closes the ambiguity that remains, so that a requirement is stated precisely enough either to be built or to be refused. The team then recommends, declining what the project should not attempt and naming the risk of attempting it; Section 2.3 records one such case.

Where the team's own understanding is put forward, it is put forward in a form that is easy to contradict: the interface is shown as a working prototype rather than described in prose, and the requirements specification sets its unresolved points out as a numbered list of questions rather than leaving them implicit. In each case the intent is to elicit correction rather than assent.

2.3 MVP Scope

Scope is prioritised by MoSCoW: **Must** is committed, **Should** is intended, **Could** is delivered only if the committed scope completes early, and **Won't** is excluded this semester. The full item-level checklist is Appendix B of the requirements specification [7]; the table below is the boundary it defines.

Function	Must	Should	Could
Map	Site markers with deployment status; site detail with the equipment and technology supplier recorded there; filter by technology supplier; time slider showing deployments in service at a chosen date	Further filters: country, region, machine type, automation level, commodity, surface or underground	
Analytics	Headline statistics; cumulative machines in service over time; counts by country and by machine type; share by technology supplier and by manufacturer; filters applying to every measure	Share by automation level and by company role; machines newly commissioned each year	User chooses which measures are displayed
Equipment register	Data model covering companies, sites, machine models, technology products and the deployments linking them; four automation levels; searchable and sortable register; export of the dataset	Machine model detail listing the deployments using it	Link from a machine model to the manufacturer's page
Contribution and verification	Submission form with immediate publication; verification status with verified records shown by default; evidence links; administration screen for review; correction of records; disclosure notice	Revision history with reversion; bulk contribution by arrangement; linked media	Sign-in with Microsoft or Google; automated evidence check

The boundary was set by need rather than by request. Both representative users asked for more than a semester allows, including a feed of market events such as acquisitions and new product launches, dashboards configurable per user, and comparison of one mine site against similar ones [1]. An item was committed as *Must* only where the client stated it and the product is incomplete without it. Requests that depend on an unresolved question, on a budget owner, or on data the project may not receive were recorded as *Should* or *Could*, or carried in the backlog, so that they remain visible without being promised.

The following are excluded this semester. A reason is recorded in each case so that the boundary is traceable [7, Sec. 7].

- **Self-service bulk upload.** Few contributors are expected to hold a full dataset, and a manual import is better value.
- **Sensor-level detail.** It is a published property of the technology product, which the system already records.
- **Joint-venture ownership percentages.** Shares are disclosed inconsistently and no requirement depends on them.
- **Fleet management and dispatch systems.** These schedule manually operated machines, so recording them would inflate adoption figures.
- **A mobile application.** Not requested.
- **Market forecasting.** Requested at the elicitation meeting, where the representative user asked whether the system could project adoption two years forward by combining the historical trend with mining companies' published forecasts. The team declined on two grounds: producing an economic projection is the work of a financial analyst rather than of this team, and a projection built on incomplete data would undermine confidence in the dataset itself. It offered instead the two capabilities an analyst needs to produce one, a time control for viewing a chosen date range and export of the dataset. The client accepted this in discussion [1].

2.4 Agreement Status and Open Questions

No part of the scope in Section 2.3 has been approved by the client. Preferences expressed by the representative users during the elicitation meeting are recorded in the transcript and the minutes, but discussion is not approval, and nothing has been confirmed in writing.

Confirmation has been pursued through every available channel. The scope was issued by email on 12 August 2026 as a checklist in which every item carries an accept or amend option, so that it can be answered without drafting a reply [7, App. B]. A message on MS Teams the same day flagged the email and identified the two answers most needed if time was short: the checklist itself, and the question about the UWA hosting environment. Follow-ups were sent by MS Teams on 14 August and by both email and Teams on 19 August, Teams being the route the client asked the team to use when email is slow [2]; the second offered a call, with an availability link, so that the checklist could be worked through together. This report is submitted into the same email thread. Every contact is recorded in the correspondence log with its date, channel and outcome.

The position was escalated within the unit. Advice was sought from the unit coordinator and the facilitator on whether the report could proceed on documented assumptions, or whether a signed agreement on the MVP scope was a requirement. The guidance was to continue following up with the client, to prepare the report without waiting, and to revise it once confirmation arrives; a one-week extension was granted on that basis. The team also verified that the delay is not particular to its own correspondence, by checking with the project manager of Group 15, which works to the same brief with the same client. Client availability was identified as a project risk in the first week rather than encountered as a surprise: the group's first meeting request, sent on 23 July, required a follow-up before it was answered. It is carried as R-01 in Section 4.

Section 2.3 is therefore a proposal awaiting validation. It is drawn from the two client meetings, the requirements elicitation with the representative users, and the written brief, and each requirement in the specification records the contact it came from [7]. What is outstanding is confirmation rather than further elicitation: a walkthrough of the checklist and the client's signature against it. Obtaining that signed agreement is the team's first priority and will be pursued as soon as contact is re-established, as directed by the unit coordinator.

Work proceeds meanwhile. Where a decision is needed and no answer has arrived, the team adopts the option proposed in the requirements specification, records it as an assumption with the date it took effect and the work it affects, and raises it at the next client contact. No committed task depends on an unanswered question, so the foundation work in Sprint 1 continues. The assumptions in force are listed below.

Question	Assumption in force	From	Work affected
Q-1a. Is the four-level automation scale acceptable?	Four levels implemented as specified: line-of-sight remote, teleremote, semi-autonomous, autonomous	22 Aug	Schema enumeration; every view that filters by automation level
Q-4. What does the UWA hosting environment provide?	Deployment plan written to accommodate each possible answer; staging serves demonstrations until the environment is known	18 Aug	Deployment approach, Section 3
Q-2, Q-3. Are accounts required to submit, and who may correct a record?	Anonymous submission accepted; any user may propose a correction, which returns the record to pending review	Sprint 3 planning	Contribution and verification
Q-7. When will seed data be available?	A documented synthetic dataset is built in Sprint 1	18 Aug	Demonstrations and analytics
Q-10. Which licences apply?	CC BY 4.0 for the dataset, MIT for the source code	Handover	Repository licensing

Each assumption is listed in the requirements specification against the requirements it affects [7, App. A], and

each is raised explicitly at the next client contact rather than left to stand by default.

3 Project Planning and Management

Delivery is organised as four two-week sprints, running from week 5 to the final report on 13 October, each concluding with a working increment deployed to staging. The GitHub repository holds the live planning artefacts, which this section summarises.

3.1 Project Management Tools

Tool	Used for	Rationale
GitHub	Source control, task assignment and tracking on a project board, sprint milestones, peer review, continuous integration and deployment, and all project documentation	Issues and milestones were adopted in preference to a separate tracking tool so that the plan, the documentation and the source share a single history and review trail. The client and facilitator can be given read access to the repository itself, so no further tool is required of them.
Mermaid	Diagrams that must remain consistent with the implementation: the entity-relationship diagram, and state or sequence diagrams where required	Maintained as code, these diagrams are versioned and reviewed within the same pull request as the change they document, so revisions are explicit to reviewers. An exported image cannot be compared between versions, and becomes outdated without detection.
draw.io	Diagrams in which the arrangement itself conveys meaning: system architecture, deployment topology and interface wireframes	Composed layout is necessary where grouping and position are the substance of the diagram, which Mermaid's automatic layout cannot express. Diagrams are exported and committed to the repository, so that no artefact depends upon a hosted account.
MS Teams	A group chat for day-to-day coordination; the unit and Group 5 channels for facilitator and academic correspondence	The client, facilitator and teaching team all work in Teams, and the unit provides the group channel, so correspondence held there is retained and searchable by default rather than dispersed across individual inboxes. Teams also hosts the online client and facilitator meetings, and its recordings and transcripts provide the basis for the minutes.

3.2 Team Workflow

- **Task assignment.** Tasks are assigned and tracked on a GitHub Projects board, each issue carrying an assignee and the milestone for its sprint [31]. A pull request stating `Closes #n` closes its issue on merge, which moves the card to Done, so board status reflects the repository rather than manual upkeep. The board is reviewed at the weekly team meeting.
- **Branching.** The team follows trunk-based development [32]: `main` is protected, always deployable and never committed to directly. Work is carried out on short-lived branches named `<type>/<issue-number>-<short-name>`, for example `feat/12-deployment-read-api`, so that a branch, an issue and a pull request denote the same unit of work. Branches are deleted once merged.
- **Commits.** Commit messages follow the Conventional Commits specification [33]: `<type>(<scope>): <description>`, for example `feat(api): add deployment read endpoint`. Squash merges append the pull request number automatically, so history records the area, the change and its origin. Branch prefixes use the same type vocabulary.
- **Review.** Branch protection requires every change to reach `main` through a pull request with at least one approving review from a member outside the relevant area and a passing CI run. A pull request template records what changed, the issue it closes and how it was tested.

- **Definition of done.** The Scrum Guide requires each team to define its own [30]. Ours: merged to `main`, CI passed, running on staging, the acceptance criteria in the issue demonstrably met, and affected documentation updated in the same pull request.
- **Meetings.** A standing weekly client meeting on Thursdays at 10:00 will be proposed at the next client contact, since frequent contact improves the fit of what is delivered. The team meeting follows at 11:00 whether or not the client meeting is taken up. The facilitator meeting is fortnightly, Tuesdays at 16:30, from 18 August.
- **Client contact.** Email in the first instance, followed by a Teams message if no response arrives within one day, as agreed on 6 August. The facilitator is copied on every client email, as he requested. Minutes are committed within 24 hours by the rotating note-taker. Where an answer is needed and none arrives, the team proceeds on a documented assumption and confirms it retrospectively.
- **Escalation.** Any item blocked for more than two working days is raised in the team channel and referred to the client or facilitator rather than absorbed by the member concerned.

`CONTRIBUTING.md` records the team working conventions in full, covering type and scope vocabularies and the issue and pull request templates.

3.3 Roles and Responsibilities

All members hold equal standing, and decisions on scope, architecture, technology and design are agreed by the team. Following the responsibility assignment matrix convention [29], each area names one accountable member, answerable for the work being completed, tested, documented and ready for integration, while any member may contribute to any area.

Code	Member	Role	Primary area
M1	Tyrone Tran (24814799)	DevOps and AI integration	Repository configuration, CI/CD, deployment environments, system integration, LLM verification and RAG pipeline
M2	Hans Lionar (24661999)	Project coordination and business analysis	Requirements, planning and task allocation, client communication, project documentation and risk register, quality assurance
M3	Zerun Zhang (24389925)	Backend development and API testing	Backend services, application logic, REST APIs, verification workflow, automated API tests
M4	Zunhao Zhang (24758673)	Database development and data integration	Entity-relationship model and migrations, PostGIS and spatial queries, data ingestion and quality, evidence metadata, database testing and performance
M5	Shee Wang (24368932)	Front-end, GIS and analytics	Public interface, interactive GIS map, analytics dashboard, submission and moderation interfaces, report interfaces
M6	Hengan Weng (24201874)	Front-end, quality assurance and RAG evaluation	Front-end support, system and accessibility testing, RAG and report-accuracy evaluation, user and administrator guides, demonstration preparation

Each member reports weekly on work completed, evidence, blockers and planned tasks. M2 compiles these into the accountability document held in the Group 5 Teams channel, which the unit coordinator, the facilitator and the teaching team can access.

`docs/accountability.md` records the detailed responsibilities and deliverables held by each member.

3.4 Milestones and Deadlines

The unit calendar places the final group report and showcase in week 12, on 13 October, leaving four sprints from week 5.

Period	Dates	Milestone
Weeks 1–4	21 Jul – 17 Aug	Completed: project allocation; client and facilitator meetings; requirements elicitation with RCT Global and Epiroc; interactive prototype; draft MVP scope sent to the client; requirements analysis and specification [7]
Sprint 1	Weeks 5–6 · 18–31 Aug	Specification and plans submitted (25 Aug). Outstanding client questions resolved, or a dated assumption recorded for each and tracked for confirmation. **Foundation:** schema, migrations, seed data, CI/CD and staging working together, shown by one mine site from the database appearing on the map
Study break	1–7 Sep	No timetabled classes. A lighter load is assumed: documentation, testing and carry-over tasks only, with no critical-path work scheduled ahead of the in-class test on 8 September
Sprint 2	Weeks 7–8 · 8–21 Sep	**Map, register and analytics:** the map completed; a user searches, filters and sorts the register, exports the dataset, and reads the headline statistics
Sprint 3	Weeks 9–10 · 22 Sep – 5 Oct	**Contribution and verification:** a contributor submits a record with evidence, and an administrator verifies, rejects or requests further evidence; deployed to the UWA environment. Software Features due 29 September, demonstration video 6 October
Sprint 4	Weeks 11–12 · 6–13 Oct	**Handover:** acceptance walkthrough with the client, defect resolution, handover documentation; final report and showcase on 13 October

Delivery began a week later than planned and this deliverable was extended to 25 August, so Sprint 1 runs in parallel with the report. After the foundation, each sprint delivers a vertical slice, a capability complete from database to interface, rather than a horizontal layer, so every increment is usable and therefore demonstrable [30]. The client is shown the increment at the end of Sprints 1, 2 and 3; the facilitator reviews progress fortnightly. Deployment to the UWA environment falls in Sprint 3 rather than last, so that the demonstration video on 6 October and the final report are both prepared against the delivered system.

`docs/project_plan.md` holds the full breakdown, including the Gantt chart and sprint task lists, and the milestones page shows current progress.

3.5 Next Deliverable: Sprint 1 Assignments

Assignments for later sprints are held in the project plan and confirmed at each sprint review, scoped so that every member owns work that is individually demonstrable by the Software Features deliverable on 29 September.

Member	Initial issue	Completion criterion
M1	CI/CD and staging pipeline	Pull requests execute linting, tests and build; `main` deploys to staging automatically
M2	Resolve outstanding client questions (four-level automation scale, UWA environment)	Each question either answered and recorded in the requirements specification, or carried as a dated assumption with the work it affects

Member	Initial issue	Completion criterion
M3	Site and deployment read APIs	Endpoints return filterable records and the API contract is documented in the repository
M4	Schema, migrations and seed data	Migrations execute without error against staging; enumerations and constraints correspond to `ERD.md`; a documented dataset loads by a single command
M5	Map markers drawn from the live API	Markers render with status colour and the supplier filter operates against stored data
M6	Manual test checklist and Sprint 1 smoke test	Checklist committed; the slice verified on staging by a member other than its author

3.6 Deployment Plan

The environments below are proposed and to be confirmed once the UWA environment is known.

Environment	Purpose	Platform	Deployed by
Local	Development	Docker Compose, comprising the application, PostgreSQL with PostGIS and seed data	Any member
Staging	Integration and client demonstration	Hosted platform as proposed in Section 4, redeployed from `main`	Continuous integration pipeline
Production	Handover	UWA-provided environment	M1, subject to manual approval

The application is containerised so that an identical image runs in all three environments, configured entirely through environment variables. Each deployment packages the application, updates the database structure, and starts the new version. If a release proves faulty, the previous version is redeployed.

3.6.1 Contingencies

The UWA hosting environment is not yet confirmed with the client. Shell access, whether Docker is permitted, whether PostGIS may be installed, outbound internet access and public reachability all remain open. Delivery proceeds while the question is unresolved, with a contingency for each outcome that would affect it materially.

- **Docker not permitted.** On a virtual machine, the application is installed from system packages behind nginx and run under systemd. If no suitable environment is available, it stays on the hosted platform and is handed over with the container image and deployment instructions.
- **Environment provisioned late.** Every increment is demonstrated on staging, so no milestone before Sprint 3 depends on the UWA environment. If it is not ready by then, deployment moves to Sprint 4; if it is still unavailable, the system is handed over for UWA to deploy later.

Neither PostGIS nor outbound access blocks delivery, since each has a documented fallback. Without PostGIS, coordinates are held as latitude and longitude columns, which serve the point rendering and bounding-box queries the map requires. Without outbound access from the server, the optional sign-in and the automated evidence check are withdrawn, both being *Could* items. No committed feature depends on outbound access: evidence links are opened by the reviewing administrator, and coordinates for new mine sites are entered during administrative review.

4 Risk, Skills and Technology Assessment

The team assessed the technology choices, available skills and project risks against the requirements of the MVP [3][7]. The aim was to use technologies the team can realistically deliver within the project timeframe while still supporting the core requirements for GIS visualisation, analytics, crowdsourced data collection and verification. Technologies unfamiliar to the team, particularly PostGIS, were identified early so that they can be prototyped before they become critical dependencies.

4.1 Technology Architecture and Justification

4.1.1 MVP Architecture

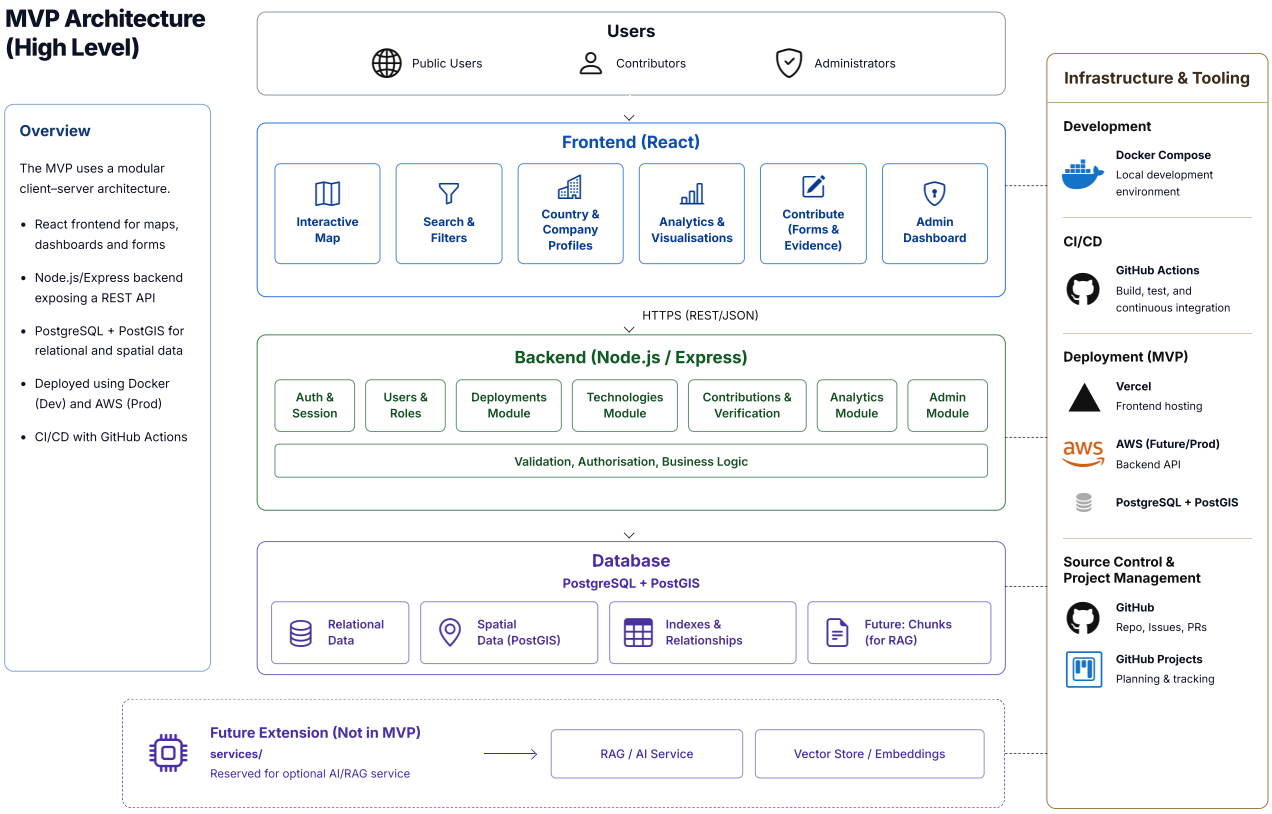


Figure 4.1: High-Level Technology Architecture for the MVP

The MVP uses a React frontend [9], a Node.js/Express backend [10][11] and a PostgreSQL/PostGIS database [12][13]. The frontend is deployed separately on Vercel [19], while the backend and database run locally under Docker Compose [17] and can later be deployed to AWS [20].

The project has one backend and a strongly related relational data model, so separating the system into independently deployed services would add integration, deployment and communication overhead without a clear benefit at this scale. The backend is instead organised into modules for deployments, contributions, administration and analytics.

The repository also contains a `services` area, reserved for optional future functionality such as AI-assisted evidence verification. RAG is not a dependency of the MVP and can be added later as a separate service without changing the core architecture [1][7].

The stack is not frozen. The facilitator confirmed that a technology choice may be revised later, provided the reason for the change and the trade-off accepted are documented in the final report [34]. The case in view is the backend host: Node.js was chosen because the members responsible for backend work are experienced in it, but the later LLM and retrieval work may be better hosted in Python. That capability sits outside the MVP, so the decision is deferred rather than taken now.

4.1.2 Technology Decisions

Decision	Alternatives considered	Chosen approach	Justification
Frontend	React selected directly based on existing team experience	**React**	At least three members already have React experience. It suits the interactive map, dashboard, filters and forms the MVP requires, and existing experience reduces implementation risk.
Backend	Python FastAPI, Node.js/Express	**Node.js + Express**	The members responsible for backend development already have Node.js/Express experience, and JavaScript can be used across both frontend and backend.
Application architecture	Microservices, modular client-server architecture	**Modular client-server architecture**	Microservices were considered but would introduce unnecessary deployment and integration complexity for the current project size. The backend can still be separated internally into clear modules.
API	REST, GraphQL	**REST with JSON**	The MVP performs CRUD, search, filtering, analytics and moderation operations. REST is straightforward to implement, test and integrate with React.
Database	MySQL, PostgreSQL	**PostgreSQL + PostGIS**	The data has strong relationships between sites, deployments, companies, equipment and technology products. PostGIS provides the geospatial functions the map uses, and PostgreSQL leaves an extension path for document chunks and vector search if RAG is implemented later.
GIS	Commercial mapping platforms, Leaflet	**Leaflet** [14]	Lightweight and open source, and sufficient for mine-site mapping without making the MVP dependent on a commercial mapping platform.
Analytics visualisation	Chart.js, Recharts	**Recharts** [15]	Integrates naturally with React and is sufficient for the required counts, market shares, category comparisons and time series.
Styling	Plain CSS, CSS frameworks	**Tailwind CSS** [16]	Supports faster development and more consistent styling across multiple frontend contributors.
Authentication	External SSO, server-managed authentication	**Session cookie authentication for the MVP**	Keeps the MVP independent of an external identity provider. Administrative requests carrying cookies also use CSRF protection.

Decision	Alternatives considered	Chosen approach	Justification
Development environment	Manual local setup, containerised environment	**Docker Compose**	Provides a reproducible environment for the backend and PostgreSQL/PostGIS and reduces differences between members' machines.
CI/CD	Manual build and test process, automated pipeline	**GitHub Actions** [18]	Integrates with the existing GitHub workflow and runs build and test checks automatically on every change submitted.
Source control and planning	Separate development and project tools	**GitHub + GitHub Projects** [31]	Keeps branches, pull requests, review, issues and task tracking in one environment already used by the team.

Hosting is proposed rather than settled. The frontend is planned for Vercel [19], and the backend and PostgreSQL/PostGIS database for local deployment under Docker with later hosted deployment on AWS [20], which together serve development and client demonstrations. Production is the UWA-provided environment described in Section 3. Both proposals remain provisional until Q-4 is answered, which asks what that environment provides and what may be installed in it [7]. Keeping the backend containerised reduces dependence on any particular platform, so redeployment into the UWA environment does not require the architecture to change.

4.2 Team Skills Assessment

The team has different strengths across frontend, backend, database, DevOps and business analysis. Existing experience covers most of the core web stack, but several areas require additional learning.

Skill required	Current capability	Gap	Action
React frontend development	At least three members have previous React experience. M5 is responsible for frontend and GIS work.	Low	Experienced members review more complex frontend work, and reusable components are shared across the project.
Node.js / Express	M1 and M3 have previous experience. M3 is responsible for the backend and API.	Low	Establish the backend structure early and use pull-request review to maintain consistency.
PostgreSQL / SQL	The team has previous SQL and database experience, and M4 is responsible for database development.	Low–Medium	Develop and test the schema early using representative synthetic data.
PostGIS	No previous project experience has been identified.	High	Build a small geospatial prototype early, work from the documentation, and have a more experienced member review the implementation before wider integration.
GIS integration	M5 holds frontend responsibility and Leaflet has been selected.	Medium	Prototype the complete flow from database coordinates → REST API → Leaflet marker before developing more advanced GIS functions.
Testing	Most members have experience with unit and system testing, mainly using Python and pytest.	Medium, in JavaScript testing tools	Transfer existing testing practice to the JavaScript stack using small examples before building the full automated suite.

Skill required	Current capability	Gap	Action
DevOps / Docker	M1 has Docker and technical integration experience.	Low	Maintain the Docker Compose configuration throughout development and test both local and hosted deployment before final delivery.
Cybersecurity	M1 has cybersecurity experience; other members have relevant course knowledge.	Medium across the wider team	Security-sensitive work is reviewed by M1, and the team works to an OWASP-based checklist during development and testing.
Accessibility	The team understands the accessibility requirements but has no dedicated specialist.	Medium	Use Lighthouse together with manual frontend checks as part of QA.
RAG / AI	M1 owns AI integration, but RAG is an identified skills gap.	High, but outside the core MVP	If the extension is undertaken, start with a small prototype and evaluate it separately before integrating it with the main application.

PostGIS is the only high-priority gap inside the MVP, and it is addressed in Sprint 1 rather than at integration: the map slice is prototyped end to end, from stored coordinates through the API to a rendered marker, before any further GIS work begins.

4.3 Resources and Constraints

Resource / constraint	Current position	Impact / response
Team capacity	Six members with defined responsibilities across project coordination, frontend, backend, database, DevOps and AI, and QA. Each member is expected to contribute at least eight hours per week outside scheduled classes and meetings	Approximately 48 person-hours per week. Work is divided by primary ownership and allocated through GitHub Projects, with peer review where skill gaps exist and reallocation when workload is uneven
Development tooling	GitHub, GitHub Projects, GitHub Actions and Docker Compose are available to the team	Supports source control, planning, automated checks and a reproducible development environment
Hosting	Vercel for the frontend, and local Docker with later AWS deployment for the backend and database	Provides a workable development and demonstration path. The UWA hosting requirements are unconfirmed pending Q-4 [7]
Budget	No budget has been confirmed. Whether funding exists for hosting or third-party services, during the project or after handover, is open question Q-4.10 [7]	The MVP uses only free or self-hosted components. A paid service will not be adopted without a confirmed budget owner, since a dependency that lapses after the semester leaves the system unusable
Development data	A documented synthetic dataset is built in Sprint 1	Allows frontend, backend, GIS and analytics work to proceed without waiting for industry data
Real seed data	The client has approached organisations including Rio Tinto and BHP, and the representative users have been asked for an export. Nothing has been received	Treated as a future resource rather than a dependency for initial MVP development. Any market-share measure remains provisional until real records exist

4.4 Risk Register

Likelihood (L) and Impact (I) are rated High, Medium or Low.

#	Risk	Category	L	I	Mitigation	Owner
R-01	Client availability delays answers to open questions or the provision of seed data, blocking dependent design and implementation decisions.	Client / Project	H	H	Consolidate blocking questions before each contact and track unresolved decisions with target dates. Where a decision cannot wait, adopt a documented and reversible working assumption and seek confirmation at the next contact. Escalate to the facilitator and unit coordinator where contact is not re-established.	M2
R-02	Real seed data is delivered late, is incomplete, or does not provide enough coverage for meaningful analytics [1][7].	Data	H	H	Develop against a documented synthetic dataset so implementation is not blocked. Replace or supplement it with verified real data when it arrives.	M4
R-03	Integration problems occur between React, Express, PostgreSQL/PostGIS and Leaflet.	Technical	H	H	Integrate vertically from an early stage using small end-to-end prototypes rather than developing each layer independently until the end.	M1
R-04	Crowdsourced records contain incorrect, incomplete or duplicate information.	Data quality	H	H	New records remain unverified, evidence is collected, administrators review records, and validation and duplicate checks are applied where possible.	M4
R-05	Analytics are read as complete market statistics although the dataset has incomplete global coverage.	Data / Business	H	H	State that measures are computed over recorded deployments, and disclose coverage limitations where they are displayed.	M2
R-06	The final deployment environment differs from the local development environment or does not provide the expected capabilities.	Technical / Deployment	M	H	Keep the backend and database containerised and test both local and hosted deployment before final delivery. Contingencies for each outcome are set out in Section 3.	M1
R-07	Uneven technical experience across members delays implementation or produces inconsistent work.	Team	H	M	Clear ownership, peer review, support from more experienced members, and small prototypes for unfamiliar technology.	M2
R-08	Additional stakeholder requests increase project scope after development has started [1][2][3].	Project	M	M	Use the proposed MVP scope and the MoSCoW priorities as the working baseline for the team until the client confirms it. New requests are assessed and recorded in the backlog before being added to committed work.	M2
R-09	The public contribution interface is abused through spam or malicious input.	Cybersecurity	L	M	CAPTCHA, rate limiting, server-side validation and controlled handling of submitted URLs.	M1

4.5 Cybersecurity Considerations

The MVP accepts public submissions, administrator actions and contributor-supplied URLs, so security controls are required even though the system is not intended to hold significant personal information.

Area	Main risk	Planned control
Authentication and authorisation	Unauthorised users reach administrative functions	Server-side session authentication and backend authorisation checks on every protected administrative endpoint [26]
CSRF	An attacker causes a signed-in administrator's browser to send an unwanted state-changing request	CSRF token required for authenticated `POST`, `PUT`, `PATCH` and `DELETE` requests [21]
SQL injection	User input changes the meaning of a database query	Parameterised queries or equivalent safe database access; no concatenation of user input into SQL [22]
XSS	Contributor content executes as JavaScript in another user's browser	Contributor text is treated as plain text and escaped on output [23]
Public submission abuse	Automated spam or excessive API requests	CAPTCHA [28], rate limiting and server-side validation
Evidence URLs and SSRF, not in the MVP	The MVP does not fetch submitted URLs; a reviewing administrator opens them in a browser. If server-side retrieval is added later for the automated evidence check (FR-55, a Could item), a submitted URL could cause the backend to reach an internal resource	Restrict URL schemes, block private and loopback addresses, and apply request timeout and response-size limits [25]
File uploads, not in the MVP	The MVP accepts evidence as links only, so no upload path is exposed. If media upload is added in a later release, malicious, unsupported or oversized files become a risk	Restrict accepted file types and sizes, validate server-side and use safe storage names [24]
Session security	Administrative session cookies are stolen or misused	`HttpOnly` and `Secure` in production, appropriate `SameSite` settings and controlled session expiry [26]
Secrets	Database credentials or application secrets are committed to GitHub	Local secrets in `.env`, excluded by `.gitignore`, and environment variables in hosted environments [27]
Dependency security	A vulnerable third-party package is introduced	Review dependencies and run build, test and security checks in the CI workflow
Credential security	Administrator passwords are exposed if credential storage is compromised	Store passwords using a strong password-hashing algorithm such as Argon2 or bcrypt; never store plaintext
Transport security	Credentials, session cookies or submitted data are intercepted in transit	HTTPS for all production browser-to-backend communication

These controls implement NFR-03 to NFR-07 of the requirements specification [7]. Administrative rules are enforced in the Node.js/Express backend rather than by hiding controls in the frontend, and security-sensitive changes are reviewed by M1.

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Acknowledgement of AI Use

Tool	Used by	Used for	How output was verified
Codex (OpenAI)	Shee Wang (24368932)	Improving the English expression of the requirements in the author's area of responsibility (M5)	The final wording was checked against the requirements specification and the entity-relationship diagram to confirm that the meaning was unchanged
Claude (Anthropic)	Hans Lionar (24661999)	Drafting and editing project documents, including the requirements specification, section 3 of this report, the project plan and the contributing guidelines; compiling meeting minutes from Teams transcripts; proofreading; explaining and checking technical concepts; locating and assessing sources	Content was checked against the client meetings, the requirements specification and the draft MVP scope, and citations against their primary sources. Errors in generated text were identified and corrected. All decisions on scope, technology and process were taken by the team, and all text was reviewed and revised before submission.
Claude Design (Anthropic)	Hans Lionar (24661999)	Producing the interactive prototype (`docs/ui/prototype.html`) from the team's static screen mockups, as an elicitation aid for the client and representative users	Theme, page content and layout were directed by the author, who reviewed the result against the agreed screen structure and corrected defects. The prototype was sent to the client so that he could test it and provide feedback.
ChatGPT (OpenAI)	Zunhao Zhang (24758673)	Reviewing the database design, identifying and aligning it with teammates' suggested changes, and assessing the potential impact of those changes	The proposed changes were independently researched and evaluated by the author and were considered appropriate. The database structure was then extended and revised in accordance with the team's suggestions.
ChatGPT (OpenAI)	Tyrone Tran (24814799)	Translating Vietnamese text into English and improving clarity and grammar while preserving the original meaning	The English translation was translated back into Vietnamese using Google Translate and compared with the original text to verify that the meaning was preserved